

LECTURE NOTES ON MULTIVARIABLE CALCULUS &
ANALYSIS

Module 1: Functions of Several Variables & Partial Differentiation —
Module 2: Differentiability & Taylor’s Theorem

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Contents

1	Module 1: Functions of Two Variables and Partial Differentiation	2
1.1	Limits and Continuity	2
1.2	Partial Derivatives	2
2	Module 2: Differentiability and Total Differentials	2
2.1	Definition of Differentiability	2
2.2	Taylor’s Theorem for Functions of Two Variables	3

1 Module 1: Functions of Two Variables and Partial Differentiation

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Multivariable calculus extends single-variable calculus concepts to functions defined on $\mathbb{R}^n$.

1.1 Limits and Continuity

Definition: Limit of a Function of Two Variables

Let $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. We say $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \varepsilon \quad (1)$$

1.2 Partial Derivatives

Definitions: Partial Derivatives

Let $f(x,y)$ be defined in a neighborhood of (a,b) .

- **Partial Derivative with respect to x at (a,b) :**

$$f_x(a,b) = \left. \frac{\partial f}{\partial x} \right|_{(a,b)} = \lim_{h \rightarrow 0} \frac{f(a+h,b) - f(a,b)}{h} \quad (2)$$

- **Partial Derivative with respect to y at (a,b) :**

$$f_y(a,b) = \left. \frac{\partial f}{\partial y} \right|_{(a,b)} = \lim_{k \rightarrow 0} \frac{f(a,b+k) - f(a,b)}{k} \quad (3)$$

2 Module 2: Differentiability and Total Differentials

2.1 Definition of Differentiability

Differentiability of $f(x,y)$ at (a,b)

A function $f(x,y)$ is said to be **differentiable** at (a,b) if the increment $\Delta f = f(a+h,b+k) - f(a,b)$ can be expressed as:

$$f(a+h,b+k) - f(a,b) = hf_x(a,b) + kf_y(a,b) + h\phi(h,k) + k\psi(h,k) \quad (4)$$

where $\phi(h,k) \rightarrow 0$ and $\psi(h,k) \rightarrow 0$ as $(h,k) \rightarrow (0,0)$.

Theorem: Differentiability Implies Continuity

Theorem: If a function $f(x, y)$ is differentiable at (a, b) , then $f(x, y)$ is continuous at (a, b) .

Proof: Taking the limit as $(h, k) \rightarrow (0, 0)$ in the differentiability increment formula yields:

$$\lim_{(h,k) \rightarrow (0,0)} f(a+h, b+k) = f(a, b)$$

Thus f is continuous at (a, b) . □

2.2 Taylor's Theorem for Functions of Two Variables**Taylor Series Expansion**

If $f(x, y)$ and its partial derivatives up to order n are continuous in a neighborhood of (a, b) , then:

$$f(a+h, b+k) = f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f(a, b) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(a, b) + \cdots + R_n \quad (5)$$